Theorem 5.3.1 : Let
- $p, s \in \mathbb{R}$ and $p \notin \mathbb{Q}$
- $b_i = \lfloor s - p(i-1) \rfloor - \lfloor s - pi \rfloor \quad i \in [\ell]$ — hence either $\lfloor p \rfloor$ or $\lceil p \rceil$
- $c_i = \{s - p(i-1)\} = s - p(i-1) - \lfloor s - p(i-1) \rfloor$
- $\sigma \in S_\ell$ s.t. $\sigma(1), \dots, \sigma(\ell)$ is in the same relative order as $c_\ell, \dots, c_1$. i.e. $\sigma(c_\ell, \dots, c_1)$ is increasing. — Put c_ℓ in the $\sigma(1)^{th}$ position, put $c_{\ell-1}$ in the $\sigma(2)^{th}$ position
- $u, v \in \mathbb{Z}_{\geq 0}$

Then

$$H_{b_1+u, b_2, b_3, \dots, b_{\ell-1}, b_\ell - v}(x) = \sum_{a_1, \dots, a_{\ell-1} \geq 0} t^{|\vec{a}|} \mathcal{L}^{\sigma}_{(b_\ell, \dots, b_1) + (-v, a_{\ell-1}, \dots, a_1)} / (a_{\ell-1}, \dots, a_1, -u)(x; q)$$

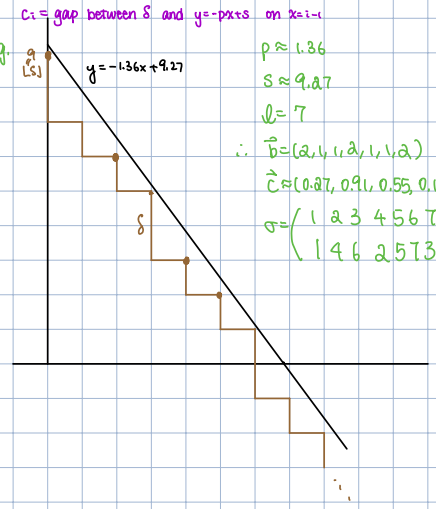
where $H^{\ell}_{\vec{b}}(x) := H^{\ell}_{q,t}\left(\dfrac{x^{\vec{b}}}{\prod_{i=1}^{\ell} (1 - qt\frac{x_i}{x_{i+1}})} \right) = H_q\left(x^{\vec{b}} \dfrac{\prod_{i+1<j} (1 - qt\frac{x_i}{x_j})}{\prod_{i<j} (1 - t\frac{x_i}{x_j})} \right)$.

Remark: If δ is the highest SE lattice path weakly below $y = -px + s$, starting at $(0, \lfloor s \rfloor)$ and extending forever along $y = 0$ (not stopping at x-axis), then

$b_i = \#$ S-steps in δ on $x = i-1$

$c_i =$ gap between δ and $y = -px + s$ on $x = i-1$

e.g.



$p \approx 1.36$

$s \approx 9.27$

$\ell = 7$

$\therefore \vec{b} = (2, 4, 1, 2, 1, 1, 2)$

$\vec{c} \approx (0.27, 0.91, 0.55, 0.19, 0.83, 0.47, 0.11)$

$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}$ (Check: $\sigma(0.11, 0.47, 0.83, 0.19, 0.55, 0.91, 0.27) = (0.11, 0.19, 0.27, 0.47, 0.55, 0.83, 0.91)$) increasing

Proof: We will prove that for $\vec{b}$ and σ as in the hypothesis of the theorem, we have the stronger 'unstraightened' identity

$$x_1^u x_\ell^{-v} x^{\vec{b}} \frac{\prod_{i+1<j} (1 - qt\frac{x_i}{x_j})}{\prod_{i<j} (1 - t\frac{x_i}{x_j})} = \sum_{a_1, \dots, a_{\ell-1} \geq 0} t^{|\vec{a}|} w_0 \left(F^{\sigma^{-1}}_{(b_\ell, \dots, b_1) + (-v, a_{\ell-1}, \dots, a_1)}(x; q) \overline{E^{\sigma^{-1}}_{(a_{\ell-1}, \dots, a_1, -u)}(x; q)} \right).$$

By Prop 4.4.2, applying H_q on both sides we will get

$$\underbrace{H_q\left(x_1^u x_\ell^{-v} x^{\vec{b}} \frac{\prod_{i+1<j} (1 - qt\frac{x_i}{x_j})}{\prod_{i<j} (1 - t\frac{x_i}{x_j})} \right)}_{H_{(b_1+u, b_2, \dots, b_{\ell-1}, b_\ell - v)}(x)} = \sum_{a_1, \dots, a_{\ell-1} \geq 0} t^{|\vec{a}|} \mathcal{L}^{\sigma}_{(b_\ell, \dots, b_1) + (-v, a_{\ell-1}, \dots, a_1)} / (a_{\ell-1}, \dots, a_1, -u)(x; q)$$

and result follows.

$\therefore$ It remains to prove the 'unstraightened' identity.

By construction, b_i takes only $\lfloor p \rfloor$ or $\lceil p \rceil$.

$\because \; b_i + c_i - c_{i+1} = \lfloor s - p(i-1) \rfloor - \lfloor s - pi \rfloor + (s - p(i-1) - \lfloor s - p(i-1) \rfloor) - (s - pi - \lfloor s - pi \rfloor) = p$

$\therefore \; b_i = \lfloor p \rfloor$ iff $c_i - c_{i+1} > 0$ iff $\sigma(\ell - i) < \sigma(\ell - i + 1)$.

Also, $b_\ell \leq b_{\ell-1} + 1$ b/c $b_\ell \leq \lceil p \rceil = \lfloor p \rfloor + 1 \leq b_{\ell-1} + 1$.

Hence $b_\ell - v \leq b_{\ell-1} - v + 1 \leq b_{\ell-1} + 1 \leq b_{\ell-1} + a_{\ell-1} + 1$, equality holds when $v + a_{\ell-1} = 0$ and $b_\ell = \lceil p \rceil$ and $b_{\ell-1} = \lfloor p \rfloor$, hence $\sigma(\ell - (\ell-1)) < \sigma(\ell - (\ell-1) + 1)$. (i.e. $\sigma(1) < \sigma(2)$)

By Lemma 4.3.4., $E^{\sigma^{-1}}_{(a_{\ell-1}, \dots, a_1, -u)}(x; q) = E^{\tau^{-1}}_{(a_{\ell-1}, \dots, a_1)}(x_1, \dots, x_{\ell-1}; q) \underbrace{E^{id}_{-u}(x_\ell; q)}_{x_\ell^{-u}} = x_\ell^{-u} E^{\tau^{-1}}_{(a_{\ell-1}, \dots, a_1)}(x_1, \dots, x_{\ell-1}; q)$ (τ: head of σ)

$$F^{\sigma^{-1}}_{(b_\ell,\dots,b_1)+(-v,a_{\ell-1},\dots,a_1)}(x;q) = \overline{E^{\sigma^{-1}\omega_0}_{(-b_\ell,\dots,-b_1)+(v,-a_{\ell-1},\dots,-a_1)}(x;q)}$$

$$= \overline{E^{(\omega_0\sigma)^{-1}}_{(-b_\ell,\dots,-b_1)+(v,-a_{\ell-1},\dots,-a_1)}(x;q)}$$

$$= \overline{E^{id}_{-b_\ell+v}(x_1;q)\, E^{(\omega_0\theta)^{-1}}_{(-b_{\ell-1},\dots,-b_1)+(-a_{\ell-1},\dots,-a_1)}(x_2,\dots,x_\ell;q)}$$

$$= \overline{x_1^{-b_\ell+v}\, E^{\theta^{-1}\omega_0}_{(-b_{\ell-1},\dots,-b_1)+(-a_{\ell-1},\dots,-a_1)}(x_2,\dots,x_\ell;q)}$$

$$= x_1^{b_\ell-v}\, F^{\theta^{-1}}_{(b_{\ell-1},\dots,b_1)+(a_{\ell-1},\dots,a_1)}(x_2,\dots,x_\ell;q)$$

Note that σ is a winding permutation of the sequence $(c_\ell,\dots,c_1)$.
Since $b_i = \lfloor p \rfloor$ iff $\sigma(\ell-i) < \sigma(\ell-i+1)$ iff $\eta_{\ell-i} = 0$, $b_{\ell-i} = \eta_i + \lfloor p \rfloor$, i.e. $(b_{\ell-1},\dots,b_1) = (\overbrace{\lfloor p \rfloor,\lfloor p \rfloor,\dots,\lfloor p \rfloor}^{\ell-1}) + \eta$.
Also $(x_1\cdots x_\ell)^k F^{\pi}_{\lambda}(x;q) = F^{\pi}_{\lambda+(\underbrace{k,\dots,k}_{\ell})}(x;q)$ for any $\pi \in S_\ell$, thus by Prop. 5.2.2,

$$F^{\sigma^{-1}}_{(b_\ell,\dots,b_1)+(-v,a_{\ell-1},\dots,a_1)}(x;q) = x_1^{b_\ell-v}\, F^{\theta^{-1}}_{(b_{\ell-1},\dots,b_1)+(a_{\ell-1},\dots,a_1)}(x_2,\dots,x_\ell;q)$$

by Prop 5.2.2

$$= x_1^{b_\ell-v}\, x_2^{\eta_1} x_3^{\eta_2}\cdots x_\ell^{\eta_{\ell-1}}\, F^{\tau^{-1}}_{(b_{\ell-1},\dots,b_1)+(a_{\ell-1},\dots,a_1)-\eta}(x_2,\dots,x_\ell;q)$$

$$= x_1^{b_\ell-v}\, x_2^{b_{\ell-1}} x_3^{b_{\ell-2}}\cdots x_\ell^{b_1}\cdot(x_2x_3\cdots x_\ell)^{-\lfloor p \rfloor}\, F^{\tau^{-1}}_{(b_{\ell-1},\dots,b_1)+(a_{\ell-1},\dots,a_1)-\eta}(x_2,\dots,x_\ell;q)$$

$$= x_1^{b_\ell-v}\, x_2^{b_{\ell-1}}\cdots x_\ell^{b_1}\, F^{\tau^{-1}}_{(b_{\ell-1},\dots,b_1)+(a_{\ell-1},\dots,a_1)\underbrace{-\eta-(\lfloor p \rfloor,\dots,\lfloor p \rfloor)}_{-(b_{\ell-1},\dots,b_1)}}(x_2,\dots,x_\ell;q)$$

$$= x_1^{b_\ell-v}\, x_2^{b_{\ell-1}}\cdots x_\ell^{b_1}\, F^{\tau^{-1}}_{(a_{\ell-1},\dots,a_1)}(x_2,\dots,x_\ell;q).$$

By Theorem 5.1.1 in $\ell-1$ variables, substituting x_i^{-1} for x_i, we get

$$\frac{\prod_{i<j}\left(1-qt\frac{y_j}{x_i}\right)}{\prod_{i\le j}\left(1-t\frac{y_j}{x_i}\right)} = \sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|}\, E^{\tau^{-1}}_{(a_1,\dots,a_{\ell-1})}(x_1^{-1},\dots,x_{\ell-1}^{-1};q^{-1})\, F^{\tau^{-1}}_{(a_1,\dots,a_{\ell-1})}(y_1,\dots,y_{\ell-1};q)$$

Set $y_j = x_{j+1}$, we get

$$\frac{\prod_{i<j}\left(1-qt\frac{x_{j+1}}{x_i}\right)}{\prod_{i\le j}\left(1-t\frac{x_{j+1}}{x_i}\right)} = \sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|}\, E^{\tau^{-1}}_{(a_1,\dots,a_{\ell-1})}(x_1^{-1},\dots,x_{\ell-1}^{-1};q^{-1})\, F^{\tau^{-1}}_{(a_1,\dots,a_{\ell-1})}(x_2,\dots,x_\ell;q)$$

$$\prod_{i<j-1}\left(1-qt\frac{x_j}{x_i}\right) = \prod_{i+1<j}\left(1-qt\frac{x_j}{x_i}\right)$$

$$\therefore\ \omega_0(x_1^u x_\ell^{-v} x^{\vec{b}})\frac{\prod_{i<j}\left(1-qt\frac{x_{j+1}}{x_i}\right)}{\prod_{i\le j}\left(1-t\frac{x_{j+1}}{x_i}\right)} = \omega_0(x_1^u x_\ell^{-v} x^{\vec{b}})\sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|}\, E^{\tau^{-1}}_{(a_1,\dots,a_{\ell-1})}(x_1^{-1},\dots,x_{\ell-1}^{-1};q^{-1})\, F^{\tau^{-1}}_{(a_1,\dots,a_{\ell-1})}(x_2,\dots,x_\ell;q)$$

$$= \sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|}\, x_\ell^u\, E^{\tau^{-1}}_{(a_1,\dots,a_{\ell-1})}(x_1^{-1},\dots,x_{\ell-1}^{-1};q^{-1})\, x_1^{b_\ell-v}\cdot x_2^{b_{\ell-1}} x_3^{b_{\ell-2}}\cdots x_\ell^{b_1}\, F^{\tau^{-1}}_{(a_1,\dots,a_{\ell-1})}(x_2,\dots,x_\ell;q)$$

$$= \sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|}\, E^{\sigma^{-1}}_{(a_{\ell-1},\dots,a_1,-u)}(x_1^{-1},\dots,x_\ell^{-1};q^{-1})\, F^{\sigma^{-1}}_{(b_\ell,\dots,b_1)+(-v,a_{\ell-1},\dots,a_1)}(x;q)$$

$$= \sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|}\, F^{\sigma^{-1}}_{(b_\ell,\dots,b_1)+(-v,a_{\ell-1},\dots,a_1)}(x;q)\, \overline{E^{\sigma^{-1}}_{(a_{\ell-1},\dots,a_1,-u)}(x;q)}$$

Applying ω_0 on both sides, we have

$$x_1^u x_\ell^{-v} x^{\vec{b}}\frac{\prod_{i+1<j}\left(1-qt\frac{x_i}{x_j}\right)}{\prod_{i\le j}\left(1-t\frac{x_i}{x_j}\right)} = \sum_{a_1,\dots,a_{\ell-1}\ge 0} t^{|\vec{a}|}\, \omega_0\left(F^{\sigma^{-1}}_{(b_\ell,\dots,b_1)+(-v,a_{\ell-1},\dots,a_1)}(x;q)\,\overline{E^{\sigma^{-1}}_{(a_{\ell-1},\dots,a_1,-u)}(x;q)}\right)$$ and result follows. $\square$